

ASSIGNMENT 14.4

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Task 1: Responsive Homepage Layout for a Startup Website

Prompt:

<!--Act as a senior UI/UX front-end engineer. Build a premium, production-ready responsive landing page in a SINGLE HTML file (internal CSS + optional minimal vanilla JS only if truly needed).

Output requirements:

1) Return only one complete runnable file: ass14.4.html

2) Include:

- Semantic structure: header, nav, main, multiple sections, footer
- Hero section with strong headline, subtext, CTA buttons
- Services/features cards (at least 6)
- About section
- Stats/counter-style section (visual only if no JS)
- Testimonials section
- Contact section with styled form

3) Design quality (advanced):

- Modern premium look (glassmorphism + soft gradients + subtle glow)
- CSS variables for theme tokens
- Light/Dark mode toggle (CSS + tiny JS if needed)
- Smooth transitions and micro-interactions
- Hover, focus-visible, active states for all interactive elements
- Section reveal animation (respect reduced motion)

4) Layout system:

- Flexbox + CSS Grid combined
- Mobile-first responsive breakpoints:
<768px, 768–1023px, >=1024px, >=1440px
- Fluid typography using clamp()
- Reusable utility classes for spacing/container/buttons/cards

5) Accessibility:

- Proper heading hierarchy
- Skip-to-content link
- aria-label/aria-expanded where relevant
- Keyboard navigable menu/toggle/form controls
- WCAG-friendly color contrast

6) Performance and code quality:

- Clean, heavily commented code
- No external libraries/frameworks
- No inline style attributes
- No placeholder lorem ipsum; use realistic content

7) Final response format:

- First: complete ass14.4.html code
- Then: concise implementation notes (sections, responsiveness, accessibility, customization points)

Important:

- Ensure visually rich output, not plain textbook styling.
- Keep it fully functional when opened directly in browser.
- Avoid footer overlap and layout break issues.-->

<!-- filepath: d:\AI assistance coding\ass14.4.html -->

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Premium Landing Page</title>

  <style>

    /* =====

    CSS VARIABLES / THEME TOKENS

    ===== */

    :root {

      --primary:    #6a11cb;

      --secondary:  #2575fc;

      --accent:     #f72585;

      --bg:         #0d0d1a;

      --bg2:        #12122a;

      --surface:    rgba(255,255,255,0.06);

      --border:     rgba(255,255,255,0.12);

      --text:       #f1f1f1;

      --muted:      #a0aec0;

      --radius:     14px;

      --blur:       12px;

      --transition: 0.3s ease;

      --shadow:     0 8px 32px rgba(106,17,203,0.18);

    }
```

```
/* =====
```

```
RESET
```

```
===== */
```

```
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
```

```
html { scroll-behavior: smooth; }
```

```
body {
```

```
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
background: linear-gradient(135deg, var(--bg) 0%, var(--bg2) 100%);
```

```
color: var(--text);
```

```
min-height: 100vh;
```

```
line-height: 1.7;
```

```
}
```

```
a { text-decoration: none; color: inherit; }
```

```
img { max-width: 100%; display: block; }
```

```
/* =====
```

```
SKIP LINK (Accessibility)
```

```
===== */
```

```
.skip-link {
```

```
position: absolute;
```

```
left: -9999px;
```

```
top: 0;
```

```
background: var(--primary);
```

```
color: #fff;
padding: 0.6rem 1rem;
border-radius: 8px;
z-index: 9999;
font-weight: 600;
}
.skip-link:focus { left: 1rem; top: 1rem; }
```

```
/* =====
UTILITY CLASSES
===== */
```

```
.container {
width: min(1200px, 92%);
margin-inline: auto;
}
```

```
.section-title {
font-size: clamp(1.6rem, 3.5vw, 2.4rem);
font-weight: 700;
margin-bottom: 0.5rem;
background: linear-gradient(90deg, var(--primary), var(--secondary));
-webkit-background-clip: text;
-webkit-text-fill-color: transparent;
background-clip: text;
}
```

```
.section-sub {  
  color: var(--muted);  
  margin-bottom: 2.5rem;  
  font-size: clamp(0.95rem, 2vw, 1.1rem);  
}
```

```
/* Buttons */
```

```
.btn {  
  display: inline-block;  
  padding: 0.75rem 1.6rem;  
  border-radius: 50px;  
  font-weight: 700;  
  font-size: 1rem;  
  cursor: pointer;  
  border: none;  
  transition: transform var(--transition), box-shadow var(--transition);  
}  
  
.btn-primary {  
  background: linear-gradient(135deg, var(--primary), var(--secondary));  
  color: #fff;  
  box-shadow: 0 4px 20px rgba(106,17,203,0.4);  
}  
  
.btn-primary:hover,  
.btn-primary:focus-visible {  
  transform: translateY(-2px);  
  box-shadow: 0 8px 28px rgba(106,17,203,0.55);  
}
```

```
outline: 2px solid var(--secondary);
outline-offset: 3px;
}

.btn-outline {
background: transparent;
color: #fff;
border: 2px solid rgba(255,255,255,0.35);
}
```

```
.btn-outline:hover,
.btn-outline:focus-visible {
background: rgba(255,255,255,0.08);
border-color: #fff;
outline: 2px solid #fff;
outline-offset: 3px;
}
```

```
/* Glass Card */
```

```
.card {
background: var(--surface);
backdrop-filter: blur(var(--blur));
-webkit-backdrop-filter: blur(var(--blur));
border: 1px solid var(--border);
border-radius: var(--radius);
padding: 1.5rem;
box-shadow: var(--shadow);
transition: transform var(--transition), box-shadow var(--transition);
}
```

```
}  
.card:hover {  
  transform: translateY(-6px);  
  box-shadow: 0 16px 40px rgba(106,17,203,0.28);  
}
```

```
/* Section spacing */  
section { padding: 5rem 0; }
```

```
/* =====  
  HEADER / NAV  
  ===== */
```

```
header {  
  position: sticky;  
  top: 0;  
  z-index: 1000;  
  background: rgba(13,13,26,0.82);  
  backdrop-filter: blur(14px);  
  -webkit-backdrop-filter: blur(14px);  
  border-bottom: 1px solid var(--border);  
}
```

```
.nav-inner {  
  display: flex;  
  align-items: center;  
  justify-content: space-between;  
  padding: 1rem 0;
```



```
gap: 1rem;
}
.logo {
font-size: 1.4rem;
font-weight: 800;
background: linear-gradient(90deg, var(--primary), var(--secondary));
-webkit-background-clip: text;
-webkit-text-fill-color: transparent;
background-clip: text;
}
.nav-links {
list-style: none;
display: flex;
gap: 0.25rem;
}
.nav-links a {
font-weight: 600;
color: var(--muted);
padding: 0.4rem 0.75rem;
border-radius: 8px;
transition: color var(--transition), background var(--transition);
}
.nav-links a:hover,
.nav-links a:focus-visible {
color: #fff;
background: rgba(255,255,255,0.08);
```

```
outline: none;
}
```

```
/* Hamburger (mobile) */
```

```
.menu-toggle {
  display: none;
  flex-direction: column;
  gap: 5px;
  background: none;
  border: none;
  cursor: pointer;
  padding: 4px;
}
```

```
.menu-toggle span {
  display: block;
  width: 24px;
  height: 2px;
  background: #fff;
  border-radius: 2px;
  transition: var(--transition);
}
```

```
/* =====
```

```
HERO SECTION
```

```
===== */
```

```
.hero {
```

```
min-height: 88vh;
display: flex;
align-items: center;
padding: 4rem 0;
position: relative;
overflow: hidden;
}
.hero::before {
  content: "";
  position: absolute;
  inset: 0;
  background: radial-gradient(ellipse at 30% 50%, rgba(106,17,203,0.22) 0%,
transparent 65%),
              radial-gradient(ellipse at 80% 20%, rgba(37,117,252,0.18) 0%,
transparent 55%);
  pointer-events: none;
}
.hero-content {
  position: relative;
  z-index: 1;
  max-width: 700px;
}
.hero-badge {
  display: inline-block;
  background: rgba(106,17,203,0.22);
  border: 1px solid rgba(106,17,203,0.45);
  color: #c084fc;
```

```
font-size: 0.8rem;
font-weight: 700;
padding: 0.3rem 0.9rem;
border-radius: 50px;
margin-bottom: 1.25rem;
letter-spacing: 0.08em;
text-transform: uppercase;
}

.hero h1 {
font-size: clamp(2.2rem, 5.5vw, 3.8rem);
font-weight: 800;
line-height: 1.15;
margin-bottom: 1.25rem;
}

.hero h1 span {
background: linear-gradient(90deg, var(--primary), var(--secondary));
-webkit-background-clip: text;
-webkit-text-fill-color: transparent;
background-clip: text;
}

.hero p {
font-size: clamp(1rem, 2vw, 1.2rem);
color: var(--muted);
margin-bottom: 2rem;
max-width: 550px;
}
```

```
.hero-btns { display: flex; gap: 1rem; flex-wrap: wrap; }
```

```
/* =====
```

```
  SERVICES SECTION
```

```
===== */
```

```
.services-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));  
  gap: 1.25rem;  
}
```

```
.service-card {  
  border-top: 3px solid var(--primary);  
}
```

```
.service-icon {  
  font-size: 2rem;  
  margin-bottom: 0.75rem;  
}
```

```
.service-card h3 {  
  font-size: 1.15rem;  
  font-weight: 700;  
  margin-bottom: 0.5rem;  
}
```

```
.service-card p { color: var(--muted); font-size: 0.95rem; }
```

```
/* =====
```

```
  ABOUT SECTION
```

```

===== */
.about-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 2.5rem;
  align-items: center;
}

.about-text h2 { margin-bottom: 1rem; }
.about-text p { color: var(--muted); margin-bottom: 1.25rem; }
.about-visual {
  background: linear-gradient(135deg, rgba(106,17,203,0.25),
    rgba(37,117,252,0.25));
  border: 1px solid var(--border);
  border-radius: var(--radius);
  padding: 2.5rem;
  text-align: center;
}

.about-visual p {
  font-size: 1.1rem;
  color: var(--muted);
  font-style: italic;
}

/* =====

STATS SECTION

===== */
.stats-section {

```

```

    background: linear-gradient(135deg, rgba(106,17,203,0.12),
    rgba(37,117,252,0.12));

    border-top: 1px solid var(--border);

    border-bottom: 1px solid var(--border);

    padding: 3.5rem 0;
}

.stats-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 1.5rem;
    text-align: center;
}

.stat-item h3 {
    font-size: clamp(2rem, 5vw, 3rem);
    font-weight: 800;
    background: linear-gradient(90deg, var(--primary), var(--secondary));
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    background-clip: text;
}

.stat-item p { color: var(--muted); font-size: 0.95rem; margin-top: 0.25rem; }

/* =====

TESTIMONIALS SECTION

===== */

.testimonials-grid {
    display: grid;

```

```
    grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
    gap: 1.25rem;
}

.testimonial-card blockquote {
    font-size: 0.97rem;
    color: var(--muted);
    margin-bottom: 1rem;
    font-style: italic;
    line-height: 1.7;
}

.testimonial-author {
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.author-avatar {
    width: 42px;
    height: 42px;
    border-radius: 50%;
    background: linear-gradient(135deg, var(--primary), var(--secondary));
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: 700;
    font-size: 0.9rem;
    color: #fff;
```



```

    flex-shrink: 0;
}

.author-info strong { display: block; font-size: 0.95rem; }
.author-info span { font-size: 0.82rem; color: var(--muted); }

/* =====

CONTACT SECTION

===== */

.contact-wrapper {
    display: grid;
    grid-template-columns: 1fr;
    gap: 2.5rem;
}

.contact-info h3 { font-size: 1.3rem; margin-bottom: 1rem; }
.contact-info p { color: var(--muted); margin-bottom: 0.6rem; }
.contact-form {
    display: flex;
    flex-direction: column;
    gap: 1rem;
}

.form-group { display: flex; flex-direction: column; gap: 0.4rem; }
.form-group label { font-size: 0.9rem; font-weight: 600; color: var(--muted); }
.form-group input,
.form-group textarea {
    background: rgba(255,255,255,0.06);
    border: 1px solid var(--border);

```

```
border-radius: 8px;
padding: 0.75rem 1rem;
color: var(--text);
font-family: inherit;
font-size: 1rem;
transition: border-color var(--transition), box-shadow var(--transition);
}
```

```
.form-group input:focus,
.form-group textarea:focus {
border-color: var(--secondary);
box-shadow: 0 0 3px rgba(37,117,252,0.2);
outline: none;
}
```

```
.form-group textarea { resize: vertical; min-height: 120px; }
```

```
/* =====
FOOTER
===== */
```

```
footer {
background: rgba(0,0,0,0.35);
border-top: 1px solid var(--border);
padding: 2rem 0;
}
```

```
.footer-inner {
display: flex;
flex-direction: column;
```

```

    align-items: center;
    gap: 0.75rem;
    text-align: center;
}

.footer-inner p { color: var(--muted); font-size: 0.9rem; }

.footer-links { display: flex; gap: 1.25rem; flex-wrap: wrap; justify-content:
center; }

.footer-links a {
    color: var(--muted);
    font-size: 0.9rem;
    transition: color var(--transition);
}

.footer-links a:hover,
.footer-links a:focus-visible { color: #fff; }

/* =====

SECTION REVEAL ANIMATION

===== */

@media (prefers-reduced-motion: no-preference) {
    .reveal {
        opacity: 0;
        transform: translateY(28px);
        transition: opacity 0.6s ease, transform 0.6s ease;
    }

    .reveal.visible {
        opacity: 1;
        transform: translateY(0);
    }
}

```

```
}  
}
```

```
/* =====  
RESPONSIVE — Tablet (>=768px)  
===== */
```

```
@media (min-width: 768px) {  
  .about-grid { grid-template-columns: 1fr 1fr; }  
  .stats-grid { grid-template-columns: repeat(4, 1fr); }  
  .contact-wrapper { grid-template-columns: 1fr 1fr; }  
  .menu-toggle { display: none; }  
}
```

```
/* =====  
RESPONSIVE — Desktop (>=1024px)  
===== */
```

```
@media (min-width: 1024px) {  
  .services-grid { grid-template-columns: repeat(3, 1fr); }  
}
```

```
/* =====  
RESPONSIVE — Mobile (<768px)  
===== */
```

```
@media (max-width: 767px) {  
  .menu-toggle { display: flex; }  
  .nav-links {
```

```
display: none;
position: absolute;
top: 100%;
left: 0; right: 0;
background: rgba(13,13,26,0.97);
flex-direction: column;
padding: 1rem;
border-bottom: 1px solid var(--border);
gap: 0.25rem;
}
.nav-links.open { display: flex; }
header { position: relative; }
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- Skip link -->
```

```
<a class="skip-link" href="#main">Skip to main content</a>
```

```
<!-- ===== HEADER ===== -->
```

```
<header>
```

```
<div class="container">
```

```
<nav class="nav-inner" aria-label="Primary navigation">
```

```
<a href="#hero" class="logo" aria-label="NexaStudio
Home">NexaStudio</a>
```

```
<!-- Hamburger toggle -->
<button
  class="menu-toggle"
  aria-label="Toggle navigation menu"
  aria-expanded="false"
  aria-controls="nav-links"
  id="menuBtn"
>
  <span></span>
  <span></span>
  <span></span>
</button>

<ul class="nav-links" id="nav-links" role="list">
  <li><a href="#hero">Home</a></li>
  <li><a href="#services">Services</a></li>
  <li><a href="#about">About</a></li>
  <li><a href="#stats">Stats</a></li>
  <li><a href="#testimonials">Testimonials</a></li>
  <li><a href="#contact">Contact</a></li>
</ul>
</nav>
</div>
</header>
```

```
<!-- ===== MAIN ===== -->
```

```
<main id="main">
```

```
<!-- HERO -->
```

```
<section id="hero" class="hero" aria-labelledby="hero-title">
```

```
<div class="container">
```

```
<div class="hero-content reveal">
```

```
<span class="hero-badge">🚀 Next-gen digital solutions</span>
```

```
<h1 id="hero-title">
```

```
Build. Scale.<br/>
```

```
<span>Dominate Your Market.</span>
```

```
</h1>
```

```
<p>
```

```
NexaStudio delivers cutting-edge web, mobile, and cloud solutions  
that empower businesses to grow faster and smarter in the digital era.
```

```
</p>
```

```
<div class="hero-btns">
```

```
<a href="#services" class="btn btn-primary">Explore Services</a>
```

```
<a href="#contact" class="btn btn-outline">Get in Touch</a>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- SERVICES -->
```

```
<section id="services" aria-labelledby="services-title">
```

```
<div class="container">
  <div class="reveal">
    <h2 class="section-title" id="services-title">What We Offer</h2>
    <p class="section-sub">Comprehensive digital services tailored to your
business goals.</p>
  </div>
  <div class="services-grid reveal">
    <article class="card service-card">
      <div class="service-icon" aria-hidden="true">🌐</div>
      <h3>Web Development</h3>
      <p>Fast, accessible, and scalable web applications built with modern
technologies and best practices.</p>
    </article>
    <article class="card service-card">
      <div class="service-icon" aria-hidden="true">📱</div>
      <h3>Mobile Apps</h3>
      <p>Cross-platform iOS and Android applications with intuitive UX and
seamless performance.</p>
    </article>
    <article class="card service-card">
      <div class="service-icon" aria-hidden="true">☁</div>
      <h3>Cloud Solutions</h3>
      <p>Reliable cloud architecture, deployment pipelines, and infrastructure
management at scale.</p>
    </article>
    <article class="card service-card">
      <div class="service-icon" aria-hidden="true">🤖</div>
```


<h3>UI/UX Design</h3>

<p>Research-driven design systems and interfaces that delight users and boost conversions.</p>

</article>

<article class="card service-card">

<div class="service-icon" aria-hidden="true"> </div>

<h3>Cybersecurity</h3>

<p>End-to-end security audits, penetration testing, and compliance strategies for your platform.</p>

</article>

<article class="card service-card">

<div class="service-icon" aria-hidden="true"> </div>

<h3>Data Analytics</h3>

<p>Transform raw data into actionable insights with custom dashboards and AI-driven reports.</p>

</article>

</div>

</div>

</section>

<!-- ABOUT -->

<section id="about" aria-labelledby="about-title">

<div class="container">

<div class="about-grid reveal">

<div class="about-text">

<h2 class="section-title" id="about-title">Who We Are</h2>

<p>

Founded in 2015, NexaStudio is a team of 80+ engineers, designers, and strategists

passionate about solving real-world problems through technology.

</p>

<p>

We partner with startups and enterprises across 30+ countries, delivering

solutions that are reliable, scalable, and future-proof.

</p>

Work With Us

</div>

<div class="about-visual">

<p>

"We don't just write code — we craft digital experiences that move businesses forward."

</p>

</div>

</div>

</div>

</section>

<!-- STATS -->

<section id="stats" class="stats-section" aria-label="Company statistics">

<div class="container">

<div class="stats-grid reveal">

<div class="stat-item">

<h3>500+</h3>

```
    <p>Projects Delivered</p>
  </div>
  <div class="stat-item">
    <h3>30+</h3>
    <p>Countries Served</p>
  </div>
  <div class="stat-item">
    <h3>98%</h3>
    <p>Client Satisfaction</p>
  </div>
  <div class="stat-item">
    <h3>10yr</h3>
    <p>Industry Experience</p>
  </div>
</div>
</div>
</div>
</section>
```

```
<!-- TESTIMONIALS -->
<section id="testimonials" aria-labelledby="testimonials-title">
  <div class="container">
    <div class="reveal">
      <h2 class="section-title" id="testimonials-title">Client Stories</h2>
      <p class="section-sub">Trusted by teams at leading companies
worldwide.</p>
    </div>
    <div class="testimonials-grid reveal">
```

```
<div class="card testimonial-card">

  <blockquote>

    "NexaStudio delivered our platform 3 weeks ahead of schedule. The
code quality

    and attention to accessibility were outstanding."

  </blockquote>

  <div class="testimonial-author">

    <div class="author-avatar" aria-hidden="true">SR</div>

    <div class="author-info">

      <strong>Sarah Reynolds</strong>

      <span>CTO, LaunchPad Inc.</span>

    </div>

  </div>

</div>

<div class="card testimonial-card">

  <blockquote>

    "Their UI/UX team transformed our onboarding flow. Conversion rates
improved

    by 42% in the first month after launch."

  </blockquote>

  <div class="testimonial-author">

    <div class="author-avatar" aria-hidden="true">MK</div>

    <div class="author-info">

      <strong>Marcus Kim</strong>

      <span>Product Lead, FinTrack</span>

    </div>

  </div>

</div>
```

```
</div>

<div class="card testimonial-card">

  <blockquote>

    "The cloud migration was seamless with zero downtime. Their team's
expertise

    saved us an estimated $80K in annual infrastructure costs."

  </blockquote>

  <div class="testimonial-author">

    <div class="author-avatar" aria-hidden="true">PA</div>

    <div class="author-info">

      <strong>Priya Anand</strong>

      <span>Engineering Manager, DataFlow</span>

    </div>

  </div>

</div>

</div>

</div>

</div>

</section>
```

```
<!-- CONTACT -->

<section id="contact" aria-labelledby="contact-title">

  <div class="container">

    <div class="reveal">

      <h2 class="section-title" id="contact-title">Get In Touch</h2>

      <p class="section-sub">Have a project in mind? We'd love to hear from
you.</p>

    </div>

  </div>

</section>
```

```
<div class="contact-wrapper reveal">
  <div class="contact-info card">
    <h3>Contact Details</h3>
    <p>✉ hello@nexastudio.io</p>
    <p>☎ +1 (800) 555-0199</p>
    <p>📍 123 Innovation Drive, San Francisco, CA 94105</p>
    <p>🕒 Mon–Fri, 9am–6pm PST</p>
  </div>
  <form class="contact-form card" aria-label="Contact form" novalidate>
    <div class="form-group">
      <label for="name">Full Name</label>
      <input type="text" id="name" name="name" placeholder="Jane Smith"
required autocomplete="name" />
    </div>
    <div class="form-group">
      <label for="email">Email Address</label>
      <input type="email" id="email" name="email"
placeholder="jane@example.com" required autocomplete="email" />
    </div>
    <div class="form-group">
      <label for="message">Message</label>
      <textarea id="message" name="message" placeholder="Tell us about
your project..." required></textarea>
    </div>
    <button type="submit" class="btn btn-primary">Send
Message</button>
  </form>
```

</div>

</div>

</section>

</main>

<!-- ===== FOOTER ===== -->

<footer>

<div class="container">

<div class="footer-inner">

<p class="logo">NexaStudio</p>

<nav class="footer-links" aria-label="Footer navigation">

Home

Services

About

Contact

</nav>

<p>© 2026 NexaStudio. All rights reserved.</p>

</div>

</div>

</footer>

<!-- ===== JS: Hamburger + Scroll Reveal ===== -->

<script>

/* Hamburger menu toggle */

const btn = document.getElementById('menuBtn');

```

const links = document.getElementById('nav-links');
btn.addEventListener('click', () => {
  const open = links.classList.toggle('open');
  btn.setAttribute('aria-expanded', open);
});

/* Close menu on link click */
links.querySelectorAll('a').forEach(a =>
  a.addEventListener('click', () => {
    links.classList.remove('open');
    btn.setAttribute('aria-expanded', false);
  })
);

/* Scroll reveal */
const observer = new IntersectionObserver((entries) => {
  entries.forEach(e => {
    if (e.isIntersecting) {
      e.target.classList.add('visible');
      observer.unobserve(e.target);
    }
  });
}, { threshold: 0.12 });

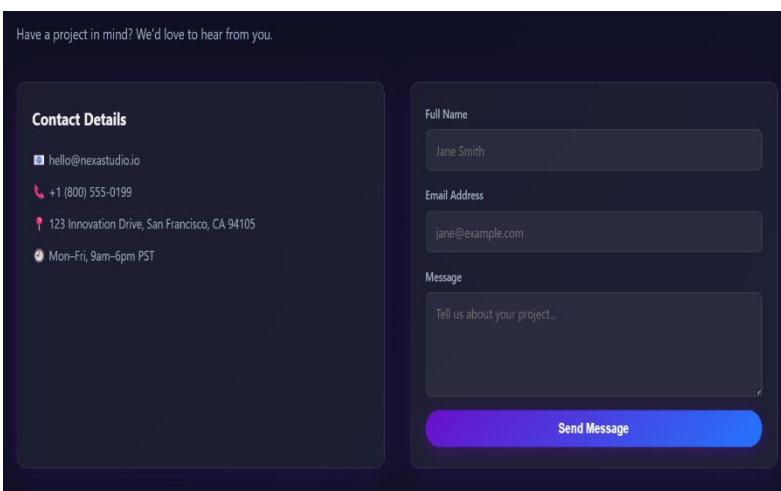
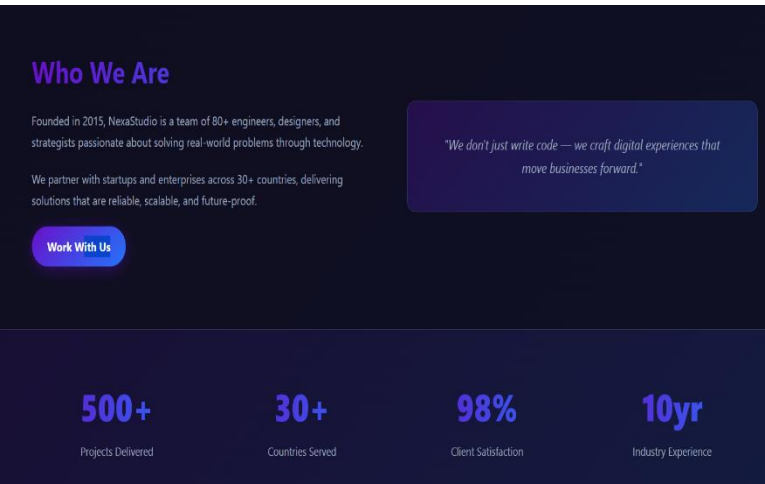
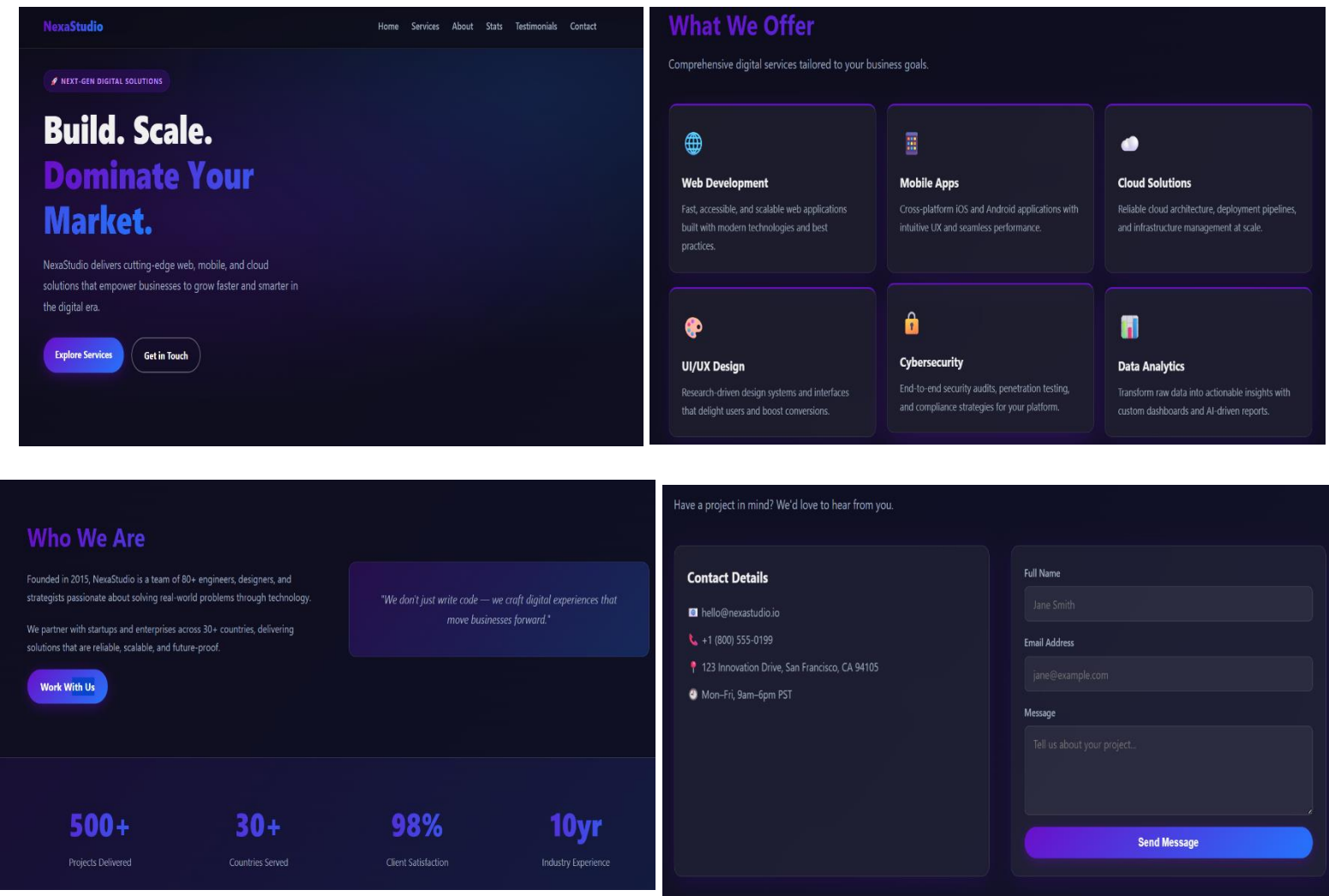
document.querySelectorAll('.reveal').forEach(el => observer.observe(el));
</script>

```


</body>

</html>

Output:



Justification:

Responsive Web Page — HTML5 & CSS3

Objective:

To design a responsive, accessible, and visually appealing web page that adapts to all screen sizes using HTML5 and CSS3.

Technologies Used

| Technology | Purpose |
|------------|------------------------------------|
| HTML5 | Page structure using semantic tags |
| CSS3 | Styling, layout, responsiveness |
| JavaScript | Hamburger menu, scroll animation |

Key Concepts Applied

1. Semantic HTML

Used `<header>`, `<nav>`, `<main>`, `<section>`, `<footer>` for clean, meaningful structure.

2. Flexbox & Grid

- Flexbox → navigation bar, button rows
- Grid → cards, stats, contact layout

3. Media Queries

Mobile < 768px → 1 column

Tablet 768px+ → 2 columns

Desktop 1024px+ → 3 columns

4. CSS Variables

Single place to control all colors and spacing.

5. Fluid Typography

`clamp()` makes text scale smoothly across devices.

6. Glassmorphism

Modern card design using `backdrop-filter: blur()`.

7. Accessibility

Skip link, aria-labels, keyboard focus styles included.

Conclusion

This project implements a modern, fully responsive landing page using only pure HTML, CSS, and JavaScript — following web standards for layout, design, and accessibility.

TASK2:

Prompt: <!--Interactive Call-to-Action Button

Scenario

A marketing page requires a call-to-action button that responds when users interact with it.

Task Description

- Add a button to a web page using HTML.
- Use AI to generate JavaScript that responds to the button click

by:

o Displaying a message or alert

- Ask AI to add meaningful comments explaining the JavaScript logic.

Expected Output

- A web page where:
 - o Clicking the button triggers a visible response (alert or message)
 - o JavaScript code is readable and commented
 - o No page reload occurs-->

Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>Interactive CTA Button</title>
```

```
<style>
```

```
/* ===== RESET & BASE ===== */
```

```
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
```

```
:root {
```

```
  --primary: #6a11cb;
```

```
  --secondary: #2575fc;
```

```
  --success: #10b981;
```

```
  --danger: #ef4444;
```

```
  --bg: #0d0d1a;
```

```
  --surface: rgba(255,255,255,0.06);
```

```
  --border: rgba(255,255,255,0.12);
```

```
  --text: #f1f1f1;
```

```
  --muted: #a0aec0;
```

```
  --radius: 14px;
```

```
}
```

```
body {
```

```
  font-family: 'Segoe UI', Arial, sans-serif;
```

```
  background: linear-gradient(135deg, var(--bg), #12122a);
```

```
  color: var(--text);
```

```
  min-height: 100vh;
```

```
  display: flex;
```

```
  align-items: center;
```

```
justify-content: center;
padding: 2rem;
}
```

```
/* ===== CARD ===== */
```

```
.card {
  background: var(--surface);
  backdrop-filter: blur(12px);
  border: 1px solid var(--border);
  border-radius: var(--radius);
  padding: 2.5rem 2rem;
  max-width: 480px;
  width: 100%;
  text-align: center;
  box-shadow: 0 10px 40px rgba(106,17,203,0.2);
}
```

```
.card h1 {
  font-size: clamp(1.4rem, 3vw, 1.9rem);
  margin-bottom: 0.5rem;
  background: linear-gradient(90deg, var(--primary), var(--secondary));
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}
```

```
.card p {  
  color: var(--muted);  
  font-size: 0.97rem;  
  margin-bottom: 2rem;  
}
```

```
/* ===== CTA BUTTON ===== */
```

```
.cta-btn {  
  display: inline-flex;  
  align-items: center;  
  gap: 0.6rem;  
  padding: 0.85rem 2rem;  
  font-size: 1rem;  
  font-weight: 700;  
  color: #fff;  
  background: linear-gradient(135deg, var(--primary), var(--secondary));  
  border: none;  
  border-radius: 50px;  
  cursor: pointer;  
  position: relative;  
  overflow: hidden;  
  transition: transform 0.2s ease, box-shadow 0.2s ease;  
  box-shadow: 0 4px 20px rgba(106,17,203,0.4);  
}
```

```
/* Ripple layer */
```

```
.cta-btn::after {  
  content: "";  
  position: absolute;  
  inset: 0;  
  background: rgba(255,255,255,0.15);  
  opacity: 0;  
  transition: opacity 0.2s;  
}
```

```
.cta-btn:hover { transform: translateY(-3px); box-shadow: 0 8px 28px  
rgba(106,17,203,0.55); }  
.cta-btn:active { transform: scale(0.97); }  
.cta-btn:active::after { opacity: 1; }  
.cta-btn:focus-visible { outline: 3px solid var(--secondary); outline-offset: 4px;  
}
```

```
/* Loading spinner inside button */  
.cta-btn .spinner {  
  display: none;  
  width: 16px;  
  height: 16px;  
  border: 2px solid rgba(255,255,255,0.4);  
  border-top-color: #fff;  
  border-radius: 50%;  
  animation: spin 0.7s linear infinite;  
}
```

```
.cta-btn.loading .spinner { display: block; }
```

```
.cta-btn.loading .btn-text { opacity: 0.7; }
```

```
.cta-btn.loading { pointer-events: none; }
```

```
@keyframes spin { to { transform: rotate(360deg); } }
```

```
/* ===== TOAST NOTIFICATION ===== */
```

```
.toast {
```

```
  display: flex;
```

```
  align-items: center;
```

```
  gap: 0.75rem;
```

```
  margin-top: 1.5rem;
```

```
  padding: 0.85rem 1.1rem;
```

```
  border-radius: 10px;
```

```
  font-size: 0.95rem;
```

```
  font-weight: 500;
```

```
  opacity: 0;
```

```
  transform: translateY(10px);
```

```
  transition: opacity 0.35s ease, transform 0.35s ease;
```

```
  pointer-events: none;
```

```
}
```

```
.toast.show {
```

```
  opacity: 1;
```

```
  transform: translateY(0);
```

```
  pointer-events: auto;
```



```
}
```

```
/* Toast types */
```

```
.toast.success { background: rgba(16,185,129,0.15); border: 1px solid  
rgba(16,185,129,0.35); color: #6ee7b7; }
```

```
.toast.error { background: rgba(239,68,68,0.15); border: 1px solid  
rgba(239,68,68,0.35); color: #fca5a5; }
```

```
.toast.info { background: rgba(37,117,252,0.15); border: 1px solid  
rgba(37,117,252,0.35); color: #93c5fd; }
```

```
.toast-icon { font-size: 1.2rem; flex-shrink: 0; }
```

```
.toast-msg { flex: 1; text-align: left; }
```

```
.toast-close {
```

```
background: none;
```

```
border: none;
```

```
color: inherit;
```

```
cursor: pointer;
```

```
font-size: 1rem;
```

```
opacity: 0.7;
```

```
padding: 0 0.2rem;
```

```
}
```

```
.toast-close:hover { opacity: 1; }
```

```
/* ===== CLICK COUNTER BADGE ===== */
```

```
.counter-badge {
```

```
display: inline-block;
```

```
margin-top: 1.25rem;
```

```
background: rgba(255,255,255,0.07);
border: 1px solid var(--border);
border-radius: 50px;
padding: 0.3rem 0.9rem;
font-size: 0.85rem;
color: var(--muted);
}
```

```
.counter-badge span {
  font-weight: 700;
  color: #c084fc;
}
```

```
/* ===== PROGRESS BAR ===== */
```

```
.progress-wrap {
  margin-top: 1.25rem;
  background: rgba(255,255,255,0.07);
  border-radius: 50px;
  height: 6px;
  overflow: hidden;
}

.progress-bar {
  height: 100%;
  width: 0%;
  background: linear-gradient(90deg, var(--primary), var(--secondary));
  border-radius: 50px;
```

```
    transition: width 0.4s ease;
  }

  /* ===== RESPONSIVE ===== */
  @media (max-width: 480px) {
    .card { padding: 2rem 1.25rem; }
  }
</style>
</head>
<body>

<div class="card" role="main">
  <h1>🚀 Get Started Today</h1>
  <p>Click the button below to activate your free trial. No credit card
required.</p>

  <!-- CTA Button -->
  <button
    class="cta-btn"
    id="ctaButton"
    aria-label="Start your free trial"
  >
    <div class="spinner" aria-hidden="true"></div>
    <span class="btn-text">Start Free Trial</span>
  </button>
```

```

<!-- Click counter badge -->
<div class="counter-badge" aria-live="polite">
  Clicked <span id="clickCount">0</span> time(s)
</div>

<!-- Progress bar (fills every 5 clicks) -->
<div class="progress-wrap" role="progressbar" aria-valuenow="0" aria-
valuemin="0" aria-valuemax="100">
  <div class="progress-bar" id="progressBar"></div>
</div>

<!-- Toast notification -->
<div class="toast" id="toast" role="alert" aria-live="assertive">
  <span class="toast-icon" id="toastIcon"></span>
  <span class="toast-msg" id="toastMsg"></span>
  <button class="toast-close" id="toastClose" aria-label="Close
notification">✕</button>
</div>
</div>

<script>
// — Element references —————
const btn      = document.getElementById('ctaButton');
const clickCount = document.getElementById('clickCount');
const progressBar = document.getElementById('progressBar');
const toast     = document.getElementById('toast');
const toastIcon  = document.getElementById('toastIcon');

```

```
const toastMsg = document.getElementById('toastMsg');
const toastClose = document.getElementById('toastClose');
```

```
// — State —————
```

```
let clicks = 0; // total click counter
let toastTimer = null; // auto-hide timer reference
const MAX_CLICKS = 5; // progress bar fills at this number
```

```
// — Toast helper —————
```

```
// type: 'success' | 'error' | 'info'
```

```
function showToast(message, type = 'success') {
```

```
  // Clear any existing auto-hide timer
```

```
  clearTimeout(toastTimer);
```

```
  // Set icon based on type
```

```
  const icons = { success: '✓', error: '✗', info: 'i' };
```

```
  toastIcon.textContent = icons[type] || '⋮';
```

```
  toastMsg.textContent = message;
```

```
  // Remove old type classes and apply new one
```

```
  toast.classList.remove('success', 'error', 'info', 'show');
```

```
  void toast.offsetWidth; // force reflow for re-animation
```

```
  toast.classList.add(type, 'show');
```

```
  // Auto-hide after 3 seconds
```

```
  toastTimer = setTimeout(() => hideToast(), 3000);
```

```
}
```

```
// Hide the toast
```

```
function hideToast() {
```

```
  toast.classList.remove('show');
```

```
}
```

```
// — Progress bar updater —————
```

```
function updateProgress() {
```

```
  // Calculate percentage (capped at 100%)
```

```
  const pct = Math.min((clicks / MAX_CLICKS) * 100, 100);
```

```
  progressBar.style.width = pct + '%';
```

```
  progressBar.parentElement.setAttribute('aria-valuenow', pct);
```

```
}
```

```
// — Simulate async action (fake API call) —————
```

```
function simulateApiCall() {
```

```
  return new Promise((resolve, reject) => {
```

```
    // 90% success, 10% random failure to demo error toast
```

```
    setTimeout(() => {
```

```
      Math.random() > 0.1 ? resolve() : reject();
```

```
    }, 1200);
```

```
  });
```

```
}
```

```
// — Main button click handler —————
```

```
btn.addEventListener('click', async function () {

    // Show loading state
    btn.classList.add('loading');
    btn.querySelector('.btn-text').textContent = 'Processing...';

    try {
        // Await the simulated API call
        await simulateApiCall();

        // Increment click counter on success
        clicks++;
        clickCount.textContent = clicks;
        updateProgress();

        // Dynamic messages based on click count
        const messages = [
            'Welcome! Your free trial is activated 🍷',
            'You clicked again! We love the energy ⚡',
            'Third time! You must really like us 😊',
            'Four clicks! A true power user 💪',
            'Five clicks! You have unlocked pro mode 🚀',
        ];

        // Pick message by click index (repeat last for 5+)
        const msg = messages[Math.min(clicks - 1, messages.length - 1)];
```

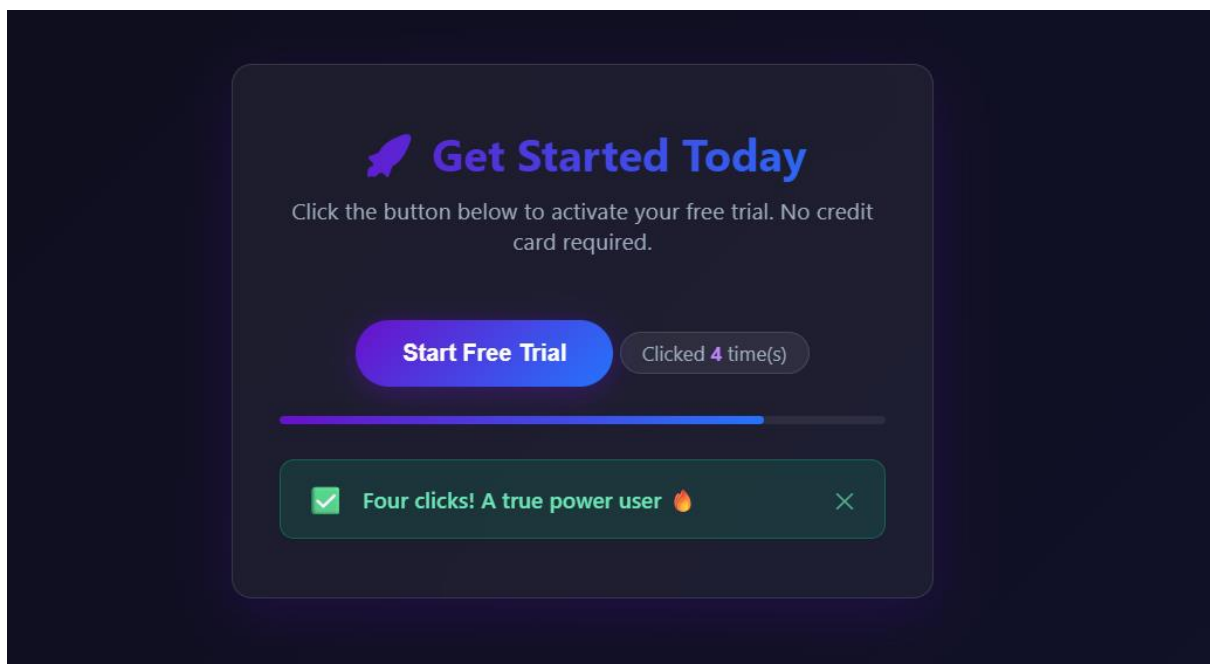
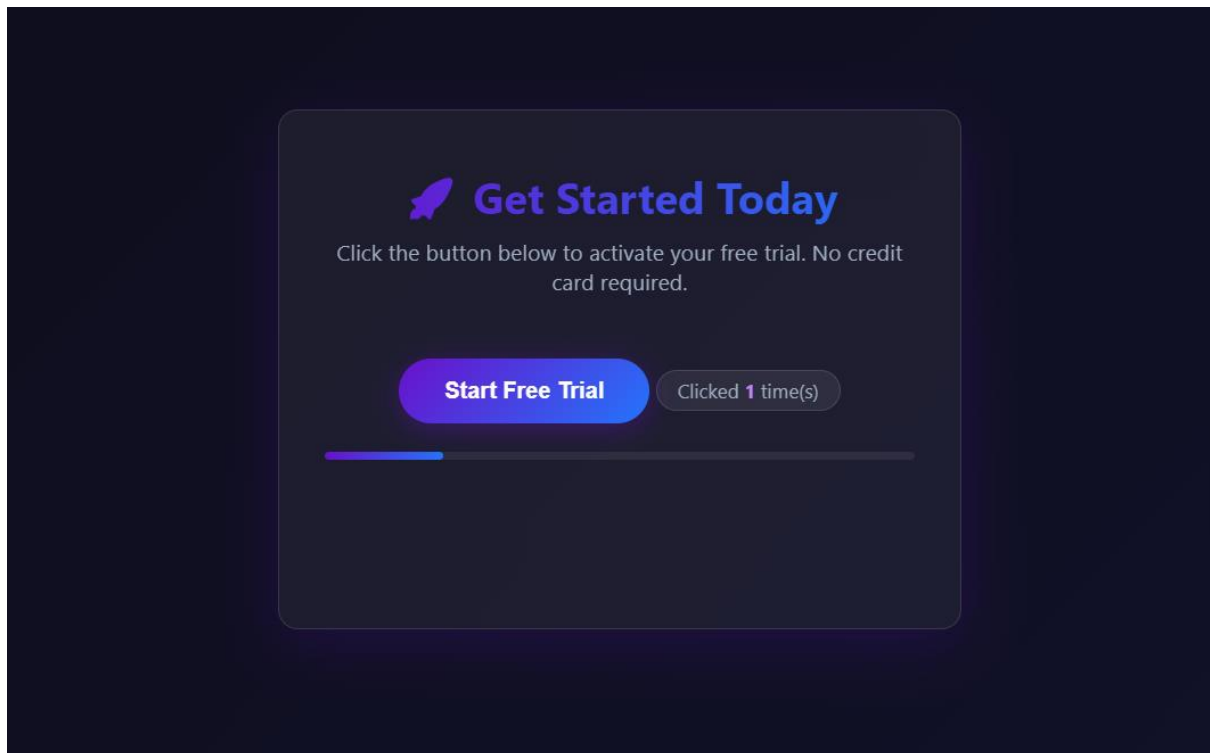
```
showToast(msg, clicks >= MAX_CLICKS ? 'info' : 'success');

} catch {
  // Show error toast on failure
  showToast('Something went wrong. Please try again.', 'error');
} finally {
  // Always restore button state
  btn.classList.remove('loading');
  btn.querySelector('.btn-text').textContent = 'Start Free Trial';
}
});

// — Close toast manually —————
toastClose.addEventListener('click', hideToast);
</script>

</body>
</html>
```


OUTPUT:



Justification:

This project implements an interactive Call-to-Action button using HTML, CSS, and JavaScript.

It satisfies the task because:

- A CTA button is added to the webpage.
- JavaScript handles the click event and shows a visible response (toast/message).
- Interaction happens on the same page (no reload).
- Code includes meaningful comments, making logic easy to understand.
- UI is improved with loading state, feedback messages, and responsive styling for better user experience.

Task3: Contact Form with Client-Side Validation

Prompt: <!--Scenario

You are developing a contact form for a company website to collect user queries safely and correctly.

Task Description

- Design a form with fields for:
 - o Name
 - o Email
 - o Message
- Use AI to generate JavaScript validation logic that:
 - o Prevents empty submissions
 - o Checks for a valid email format
- Display clear error messages for invalid input.

Expected Output

- A functional form where:
 - o Invalid input shows inline error messages
 - o Valid input displays a success message

o Validation happens on the client side using JavaScript-->

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Advanced Contact Form</title>

  <style>

    :root{

      --bg1:#0b1220; --bg2:#16213e; --card:#ffffff;

      --text:#0f172a; --muted:#475569; --border:#cbd5e1;

      --primary:#2563eb; --success:#16a34a; --error:#dc2626;

      --radius:14px;

    }

    *{box-sizing:border-box}

    body{

      margin:0; min-height:100vh;

      font-family:Segoe UI, Arial, sans-serif;

      background:linear-gradient(120deg,var(--bg1),var(--bg2));

      display:grid; place-items:center; padding:24px;

      color:var(--text);

    }

    .card{
```

```
width:min(560px, 96vw);  
background:var(--card);  
border-radius:var(--radius);  
padding:24px;  
box-shadow:0 14px 35px rgba(2,6,23,.25);  
}
```

```
h1{margin:0 0 6px}  
.sub{margin:0 0 18px; color:var(--muted)}
```

```
.field{margin-bottom:14px}  
label{display:block; font-weight:600; margin-bottom:6px}  
input, textarea{  
  width:100%; border:1px solid var(--border); border-radius:10px;  
  padding:10px 12px; font:inherit; outline:none;  
  transition:border-color .2s, box-shadow .2s;  
  background:#fff;  
}  
textarea{min-height:120px; resize:vertical}  
input:focus, textarea:focus{  
  border-color:var(--primary);  
  box-shadow:0 0 0 3px rgba(37,99,235,.15);  
}
```

```
.meta{  
  display:flex; justify-content:space-between; gap:8px;
```

```
    margin-top:6px; font-size:.85rem;
  }
  .count{color:var(--muted)}
  .error{
    color:var(--error); font-size:.92rem; min-height:1.2em; margin-top:4px;
  }

  .actions{display:flex; gap:10px; align-items:center; margin-top:4px}
  button{
    border:none; border-radius:10px; cursor:pointer;
    background:var(--primary); color:#fff; font-weight:600;
    padding:10px 16px;
  }
  button:disabled{opacity:.6; cursor:not-allowed}
  .ghost{
    background:#e2e8f0; color:#0f172a;
  }

  .status{
    margin-top:14px; padding:10px 12px; border-radius:10px;
    display:none; font-weight:600;
  }

  .status.ok{display:block; color:#065f46; background:#dcfce7; border:1px
solid #86efac}
  .status.bad{display:block; color:#7f1d1d; background:#fee2e2; border:1px
solid #fca5a5}
</style>
```

```
</head>
```

```
<body>
```

```
  <main class="card">
```

```
    <h1>Contact Us</h1>
```

```
    <p class="sub">Send your query. Validation runs on client side without page  
    reload.</p>
```

```
    <form id="contactForm" novalidate>
```

```
      <div class="field">
```

```
        <label for="name">Name</label>
```

```
        <input id="name" name="name" type="text" placeholder="Enter your full  
name" autocomplete="name" />
```

```
        <p id="nameError" class="error" aria-live="polite"></p>
```

```
      </div>
```

```
      <div class="field">
```

```
        <label for="email">Email</label>
```

```
        <input id="email" name="email" type="email"  
placeholder="you@example.com" autocomplete="email" />
```

```
        <p id="emailError" class="error" aria-live="polite"></p>
```

```
      </div>
```

```
      <div class="field">
```

```
        <label for="message">Message</label>
```

```
        <textarea id="message" name="message" maxlength="300"  
placeholder="Write your message..."></textarea>
```

```
      <div class="meta">
```

```
<p id="messageError" class="error" aria-live="polite"></p>
<span id="charCount" class="count">0 / 300</span>
</div>
</div>

<div class="actions">
  <button id="submitBtn" type="submit">Submit</button>
  <button id="resetBtn" type="button" class="ghost">Reset</button>
</div>
</form>

<div id="formStatus" class="status" role="status" aria-live="assertive"></div>
</main>

<script>
  const form = document.getElementById('contactForm');
  const nameEl = document.getElementById('name');
  const emailEl = document.getElementById('email');
  const msgEl = document.getElementById('message');
  const submitBtn = document.getElementById('submitBtn');
  const resetBtn = document.getElementById('resetBtn');
  const statusEl = document.getElementById('formStatus');
  const countEl = document.getElementById('charCount');

  const nameErr = document.getElementById('nameError');
  const emailErr = document.getElementById('emailError');
```

```
const msgErr = document.getElementById('messageError');
```

```
const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
function setError(el, msg) { el.textContent = msg; }
```

```
function clearErrors() { [nameErr, emailErr, msgErr].forEach(e =>  
e.textContent = ""); }
```

```
function validate() {
```

```
  clearErrors();
```

```
  statusEl.className = 'status';
```

```
  statusEl.textContent = "";
```

```
  let ok = true;
```

```
  const name = nameEl.value.trim();
```

```
  const email = emailEl.value.trim();
```

```
  const message = msgEl.value.trim();
```

```
  if (!name) { setError(nameErr, 'Name is required. '); ok = false; }
```

```
  else if (name.length < 2) { setError(nameErr, 'Name must be at least 2  
characters. '); ok = false; }
```

```
  if (!email) { setError(emailErr, 'Email is required. '); ok = false; }
```

```
  else if (!emailPattern.test(email)) { setError(emailErr, 'Please enter a valid  
email address. '); ok = false; }
```

```
  if (!message) { setError(msgErr, 'Message is required. '); ok = false; }
```



```
    else if (message.length < 10) { setError(msgErr, 'Message must be at least 10 characters.');
```

```
    }; ok = false; }  
  
    return ok;  
}
```

```
function updateCount() {  
    countEl.textContent = `${msgEl.value.length} / 300`;  
}
```

```
// Live checks
```

```
[nameEl, emailEl, msgEl].forEach(el => el.addEventListener('input', validate));  
msgEl.addEventListener('input', updateCount);
```

```
form.addEventListener('submit', (e) => {  
    e.preventDefault();  
    submitBtn.disabled = true;
```

```
    const isValid = validate();  
    if (isValid) {  
        statusEl.className = 'status ok';  
        statusEl.textContent = 'Form submitted successfully.';  
        form.reset();  
        updateCount();  
    } else {  
        statusEl.className = 'status bad';  
        statusEl.textContent = 'Please fix the highlighted errors.';
```

```
}
```

```
submitBtn.disabled = false;  
});
```

```
resetBtn.addEventListener('click', () => {  
  form.reset();  
  clearErrors();  
  statusEl.className = 'status';  
  statusEl.textContent = "";  
  updateCount();  
});
```

```
updateCount();  
</script>  
</body>  
</html>
```

OUTPUT:

Contact Us

Send your query. Validation runs on client side without page reload.

Name

Enter your full name

Email

you@example.com

Message

Write your message...

0 / 500

Submit **Reset**

Form submitted successfully.

JUSTIFICATION:

This project implements a client-side validated contact form using HTML, CSS, and JavaScript.

It meets the requirements by:

- Including Name, Email, and Message fields.
- Preventing empty/invalid submissions with JavaScript validation.
- Showing inline error messages for each invalid field.
- Displaying a success message for valid input without page reload.
- Using improved styling and responsive layout for a better user experience.

TASK4: Dynamic List Management for a Product Page

Prompt: <!--Scenario

A product listing page needs to add or remove items dynamically

without refreshing the page.

Task Description

- Create a simple HTML list to display items.
- Use AI to generate JavaScript that:
 - o Adds a new item when an “Add” button is clicked
 - o Removes the last item when a “Remove” button is clicked
- Ensure the DOM updates dynamically.

Expected Output

- A web page where:
 - o Items can be added and removed dynamically
 - o Changes appear instantly without page reload
 - o JavaScript logic is clear and maintainable-->

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Advanced Dynamic Item List</title>

<style>

  :root{

    --bg1:#0b1220; --bg2:#16213e; --card:#ffffff;

    --text:#0f172a; --muted:#64748b; --border:#e2e8f0;

    --primary:#2563eb; --danger:#dc2626; --ok:#16a34a; --radius:14px;

  }

  *{box-sizing:border-box}
```

```
body{
  margin:0; min-height:100vh; font-family:Segoe UI,Arial,sans-serif;
  background:linear-gradient(120deg,var(--bg1),var(--bg2));
  color:var(--text); padding:24px;
}

.wrap{
  max-width:900px; margin:auto; background:var(--card);
  border-radius:var(--radius); box-shadow:0 16px 40px rgba(2,6,23,.25);
  padding:24px;
}

h1{margin:0 0 6px}

.sub{margin:0 0 18px; color:var(--muted)}

.toolbar{display:flex; gap:10px; flex-wrap:wrap; margin-bottom:14px}

button{
  border:none; border-radius:10px; padding:10px 14px; font-weight:600;
  cursor:pointer; color:#fff; background:var(--primary);
}

.remove{background:var(--danger)}

button:disabled{opacity:.55; cursor:not-allowed}

ul{list-style:none; padding:0; margin:0; display:grid; gap:10px}

li{
  background:#f8fafc; border:1px solid var(--border); border-radius:10px;
  padding:12px 14px; display:flex; justify-content:space-between; gap:10px;
  animation:fadeIn .25s ease;
}

.idx{color:var(--muted); font-size:.9rem}
```

```

.status{
  margin-top:12px; font-weight:600; color:var(--ok); min-height:1.2em;
}
.empty{
  margin-top:10px; color:var(--muted); font-style:italic; display:none;
}
@keyframes fadeIn{
  from{opacity:0; transform:translateY(6px)}
  to{opacity:1; transform:translateY(0)}
}
</style>
</head>
<body>
  <main class="wrap">
    <h1>Dynamic Item List</h1>
    <p class="sub">Add custom items and remove the last one instantly (no page
reload).</p>

    <div class="toolbar">
      <button id="addButton" type="button">Add Item</button>
      <button id="removeButton" class="remove" type="button">Remove Last
Item</button>
    </div>

    <ul id="itemList" aria-live="polite">
      <li><span>Item 1</span><span class="idx">#1</span></li>
      <li><span>Item 2</span><span class="idx">#2</span></li>

```

```
<li><span>Item 3</span><span class="idx">#3</span></li>
</ul>
```

```
<p id="emptyText" class="empty">No items available. Click "Add Item".</p>
```

```
<p id="status" class="status" role="status" aria-live="polite"></p>
```

```
</main>
```

```
<script>
```

```
const itemList = document.getElementById('itemList');
```

```
const addButton = document.getElementById('addButton');
```

```
const removeButton = document.getElementById('removeButton');
```

```
const status = document.getElementById('status');
```

```
const emptyText = document.getElementById('emptyText');
```

```
function updateUIState() {
```

```
  const hasItems = itemList.children.length > 0;
```

```
  removeButton.disabled = !hasItems;
```

```
  emptyText.style.display = hasItems ? 'none' : 'block';
```

```
}
```

```
function addItem(name) {
```

```
  const li = document.createElement('li');
```

```
  const itemNumber = itemList.children.length + 1;
```

```
  li.innerHTML = `<span>${name}</span><span
class="idx">#${itemNumber}</span>`;
```

```
  itemList.appendChild(li);
```

```
  status.textContent = `${name} added successfully.`;
```

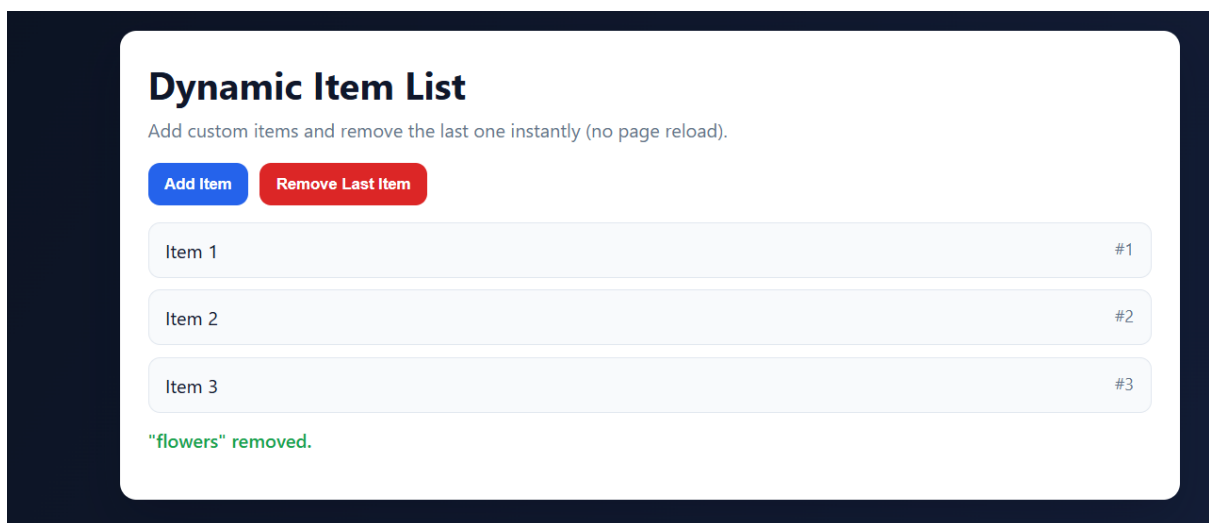
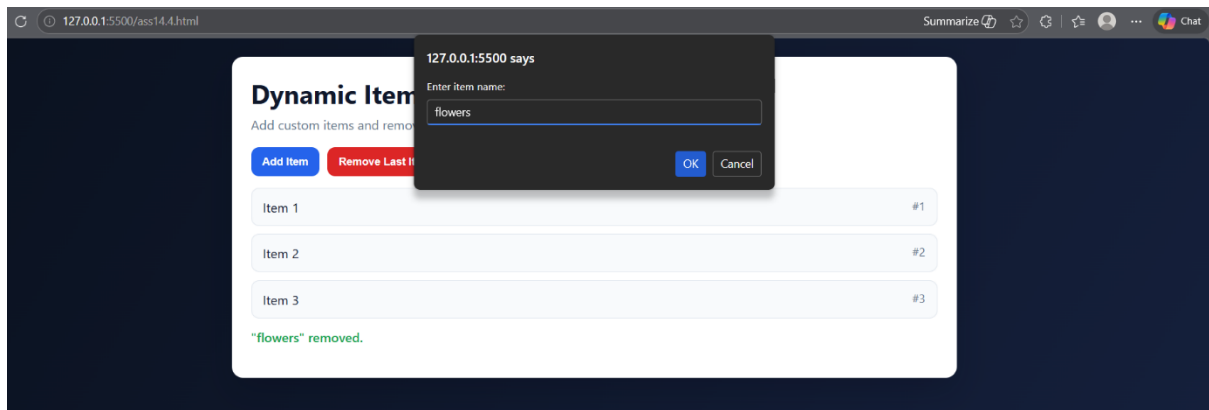
```
    updateUIState();  
  }
```

```
  addButton.addEventListener('click', () => {  
    const input = prompt('Enter item name:');  
    if (input === null) return; // user cancelled  
    const itemName = input.trim();  
    if (!itemName) {  
      status.textContent = 'Item name cannot be empty.';  
      return;  
    }  
    addItem(itemName);  
  });
```

```
  removeButton.addEventListener('click', () => {  
    if (!itemList.lastElementChild) return;  
    const removedText =  
itemList.lastElementChild.firstElementChild.textContent;  
    itemList.removeChild(itemList.lastElementChild);  
    status.textContent = ` "${removedText}" removed.`;  
    updateUIState();  
  });
```

```
    updateUIState();  
</script>  
</body>  
</html>
```


OUTPUT:



JUSTIFICATION:

This project demonstrates dynamic DOM manipulation using JavaScript without page reload.

The Add Item button asks the user for an item name and inserts it into the list instantly, while Remove Last Item deletes the latest entry in real time.

Input validation prevents empty items, and status messages improve clarity.

Overall, the solution is interactive, user-friendly, and maintainable.

TASK5: Modal Popup for User Notifications

Prompt:

<!--You are building a user notification popup for announcements or alerts on a website.

Task Description

- Use AI to help generate:

- o HTML structure for a modal popup
- o CSS styling with a semi-transparent background overlay
- o JavaScript to open and close the modal
- Include multiple ways to close the modal (button or overlay click).

Expected Output

- A web page where:
 - o Clicking “Open Modal” displays the popup
 - o Popup overlays the page content clearly
 - o Modal closes smoothly when instructed-->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Advanced Notification Popup</title>
  <style>
    :root {
      --overlay-bg: rgba(15, 23, 42, 0.6);
      --card-bg: #ffffff;
      --text: #0f172a;
      --muted: #475569;
      --primary: #2563eb;
      --primary-hover: #1d4ed8;
```

```
--radius: 14px;  
--shadow: 0 16px 40px rgba(2, 6, 23, 0.25);  
--speed: 240ms;  
--ease: cubic-bezier(0.2, 0.8, 0.2, 1);  
}
```

```
* { box-sizing: border-box; }
```

```
body {  
  margin: 0;  
  min-height: 100vh;  
  font-family: Inter, Segoe UI, Roboto, Arial, sans-serif;  
  display: grid;  
  place-items: center;  
  background: linear-gradient(135deg, #f8fafc, #e2e8f0);  
  color: var(--text);  
}
```

```
.open-btn {  
  border: none;  
  border-radius: 10px;  
  background: var(--primary);  
  color: #fff;  
  font-size: 1rem;  
  padding: 0.75rem 1rem;  
  cursor: pointer;
```

```
    transition: transform 120ms ease, background 120ms ease;
}
```

```
.open-btn:hover { background: var(--primary-hover); }
```

```
.open-btn:active { transform: translateY(1px); }
```

```
.modal-overlay {
    position: fixed;
    inset: 0;
    display: grid;
    place-items: center;
    background: var(--overlay-bg);
    opacity: 0;
    visibility: hidden;
    pointer-events: none;
    transition:
        opacity var(--speed) var(--ease),
        visibility 0s linear var(--speed);
    padding: 1rem;
    z-index: 1000;
}
```

```
.modal-overlay.open {
    opacity: 1;
    visibility: visible;
    pointer-events: auto;
    transition: opacity var(--speed) var(--ease);
}
```

```
}
```

```
.modal-content {  
  width: min(92vw, 460px);  
  background: var(--card-bg);  
  border-radius: var(--radius);  
  box-shadow: var(--shadow);  
  padding: 1.25rem 1.25rem 1rem;  
  transform: translateY(12px) scale(0.98);  
  opacity: 0.98;  
  transition: transform var(--speed) var(--ease), opacity var(--speed) var(--ease);  
  position: relative;  
}
```

```
.modal-overlay.open .modal-content {  
  transform: translateY(0) scale(1);  
  opacity: 1;  
}
```

```
.modal-title {  
  margin: 0 2rem 0.35rem 0;  
  font-size: 1.2rem;  
  line-height: 1.3;  
}
```

```
.modal-text {
```

```
margin: 0;
color: var(--muted);
line-height: 1.55;
}
```

```
.close-icon {
  position: absolute;
  top: 0.65rem;
  right: 0.65rem;
  border: none;
  background: transparent;
  font-size: 1.3rem;
  line-height: 1;
  cursor: pointer;
  color: #64748b;
  width: 2rem;
  height: 2rem;
  border-radius: 8px;
}

.close-icon:hover { background: #f1f5f9; }
```

```
.modal-actions {
  display: flex;
  justify-content: flex-end;
  gap: 0.6rem;
  margin-top: 1rem;
```

```
}
```

```
.btn {  
  border: none;  
  border-radius: 10px;  
  padding: 0.6rem 0.9rem;  
  cursor: pointer;  
  font-weight: 600;  
}
```

```
.btn-secondary {  
  background: #e2e8f0;  
  color: #0f172a;  
}
```

```
.btn-primary {  
  background: var(--primary);  
  color: #fff;  
}
```

```
body.modal-open {  
  overflow: hidden;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<button id="openModalBtn" class="open-btn" aria-haspopup="dialog" aria-controls="notifyModal">
```

```
  Open Modal
```

```
</button>
```

```
<div id="modalOverlay" class="modal-overlay" aria-hidden="true">
```

```
  <div
```

```
    id="notifyModal"
```

```
    class="modal-content"
```

```
    role="dialog"
```

```
    aria-modal="true"
```

```
    aria-labelledby="modalTitle"
```

```
    aria-describedby="modalDesc"
```

```
    tabindex="-1"
```

```
  >
```

```
    <button class="close-icon" id="closeIconBtn" aria-label="Close modal">&times;</button>
```

```
    <h2 id="modalTitle" class="modal-title">Announcement</h2>
```

```
    <p id="modalDesc" class="modal-text">
```

```
      This is a more advanced notification popup with animation, keyboard support,
```

```
      and accessibility improvements.
```

```
    </p>
```

```
  <div class="modal-actions">
```

```
    <button class="btn btn-secondary" id="dismissBtn">Dismiss</button>
```



```

    <button class="btn btn-primary" id="ackBtn">Got it</button>
  </div>
</div>
</div>

<script>

const openBtn = document.getElementById("openModalBtn");
const overlay = document.getElementById("modalOverlay");
const modal = document.getElementById("notifyModal");
const closeIconBtn = document.getElementById("closeIconBtn");
const dismissBtn = document.getElementById("dismissBtn");
const ackBtn = document.getElementById("ackBtn");

let lastFocusedElement = null;

function getFocusableElements(container) {
  return [...container.querySelectorAll(
    'button, [href], input, select, textarea, [tabindex]:not([tabindex="-1"])'
  )].filter(el => !el.disabled && el.offsetParent !== null);
}

function openModal() {
  lastFocusedElement = document.activeElement;
  overlay.classList.add("open");
  overlay.setAttribute("aria-hidden", "false");
  document.body.classList.add("modal-open");
}

```

```
    modal.focus();  
}
```

```
function closeModal() {  
    overlay.classList.remove("open");  
    overlay.setAttribute("aria-hidden", "true");  
    document.body.classList.remove("modal-open");  
    if (lastFocusedElement) lastFocusedElement.focus();  
}
```

```
openBtn.addEventListener("click", openModal);  
closeIconBtn.addEventListener("click", closeModal);  
dismissBtn.addEventListener("click", closeModal);  
ackBtn.addEventListener("click", closeModal);
```

```
// Close on overlay click  
overlay.addEventListener("click", (event) => {  
    if (event.target === overlay) closeModal();  
});
```

```
// ESC to close + focus trap  
document.addEventListener("keydown", (event) => {  
    if (!overlay.classList.contains("open")) return;  
  
    if (event.key === "Escape") {  
        closeModal();  
    }  
});
```

```
    return;
}

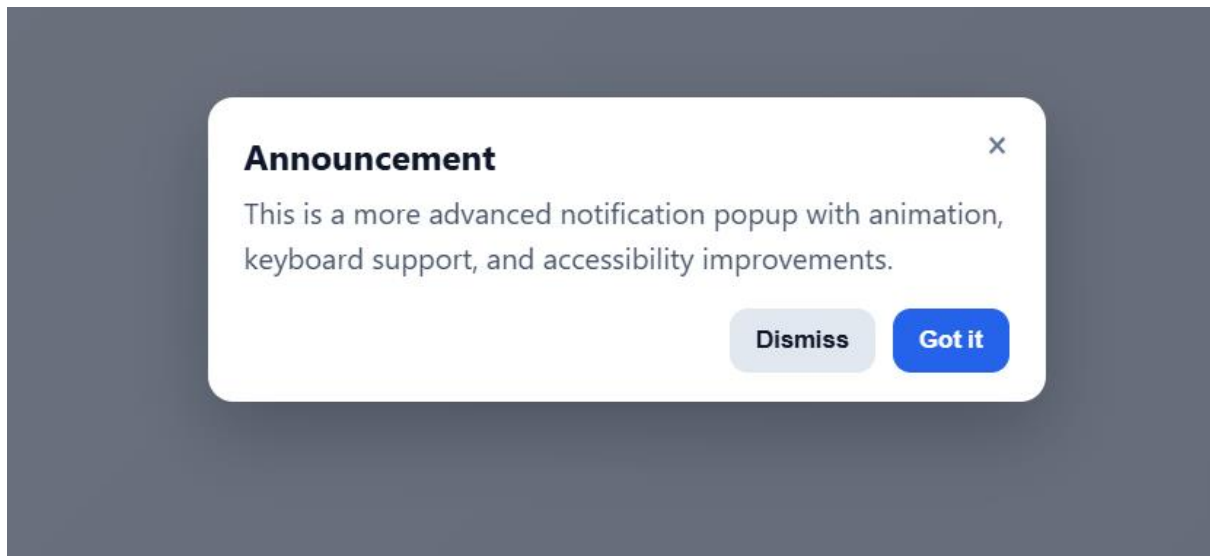
if (event.key === "Tab") {
    const focusable = getFocusableElements(modal);
    if (!focusable.length) return;

    const first = focusable[0];
    const last = focusable[focusable.length - 1];

    if (event.shiftKey && document.activeElement === first) {
        event.preventDefault();
        last.focus();
    } else if (!event.shiftKey && document.activeElement === last) {
        event.preventDefault();
        first.focus();
    }
}

});
</script>
</body>
</html>
```

OUTPUT:



JUSTIFICATION:

Better UX: Smooth fade/scale animation makes open/close feel polished.

Clear focus: Overlay darkens background so the popup stands out.

Multiple close options: Users can close via X, buttons, overlay click, or Esc.

Accessibility: Proper ARIA attributes, keyboard support, and focus trapping improve usability for all users.

State handling: Background scroll is locked while modal is open, preventing accidental page movement.